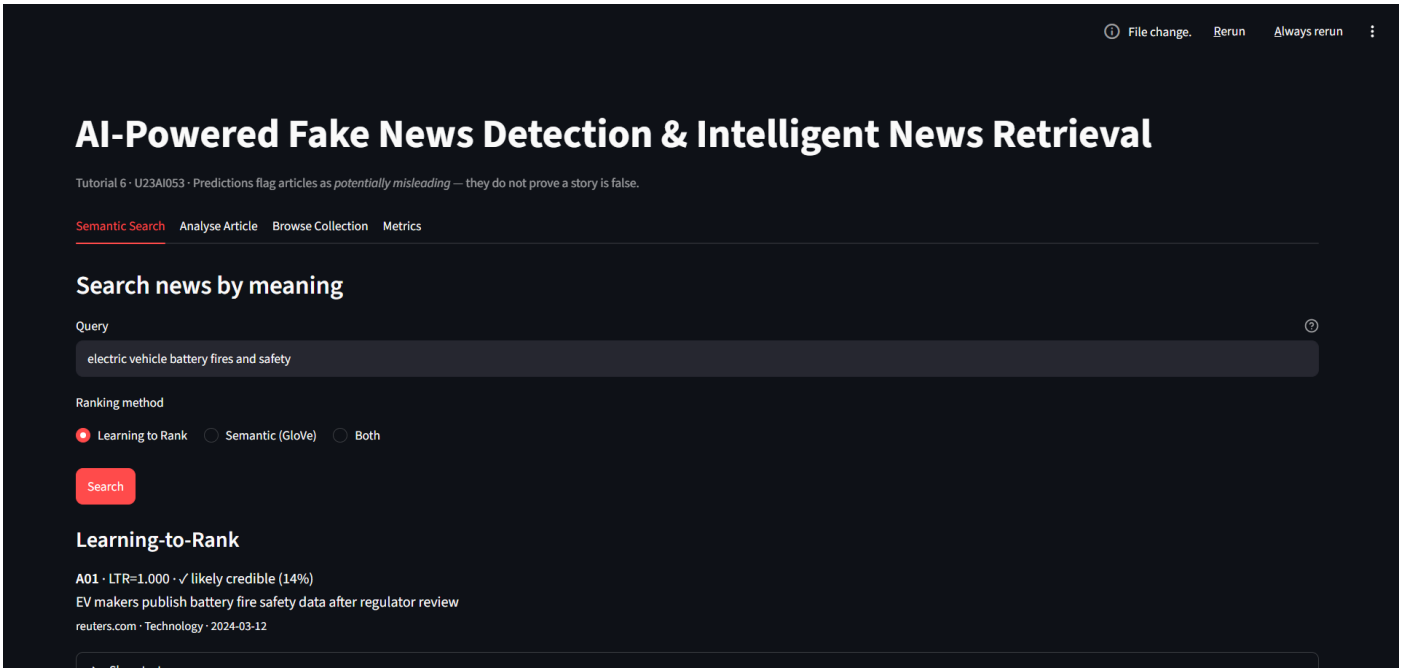
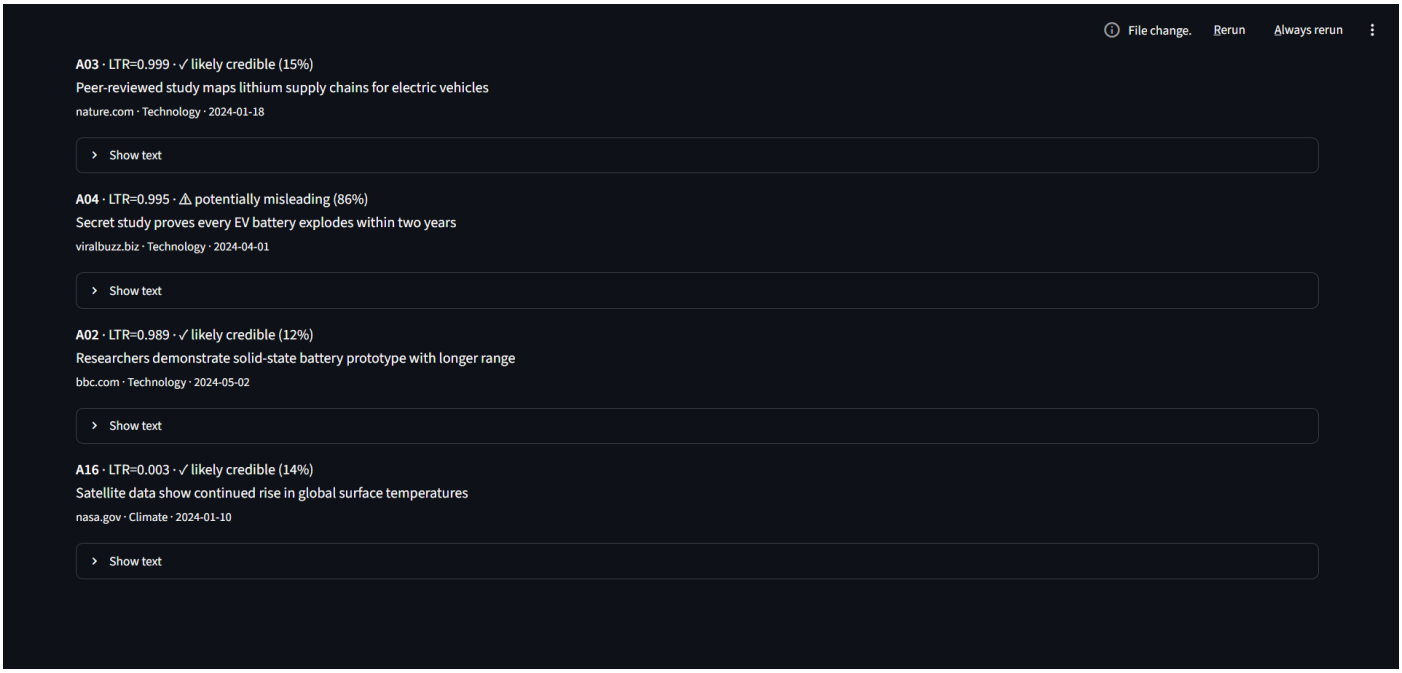
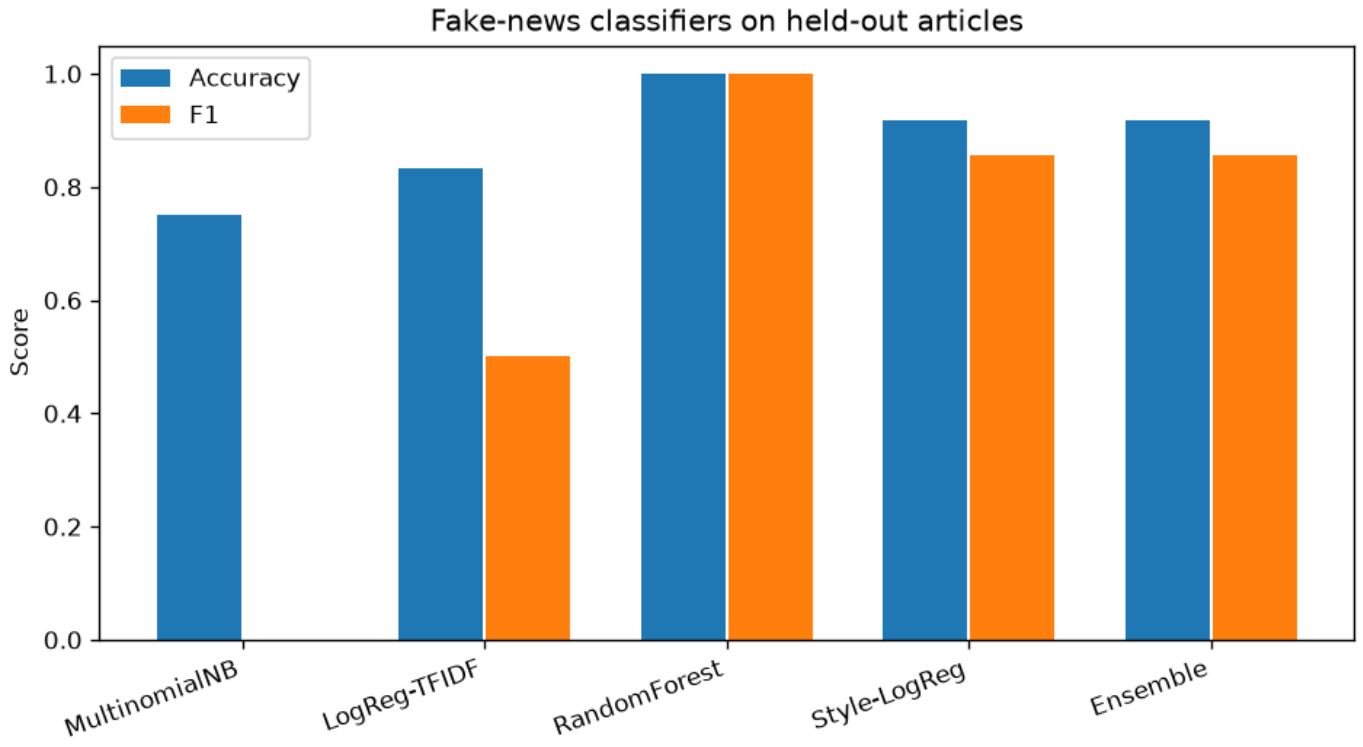
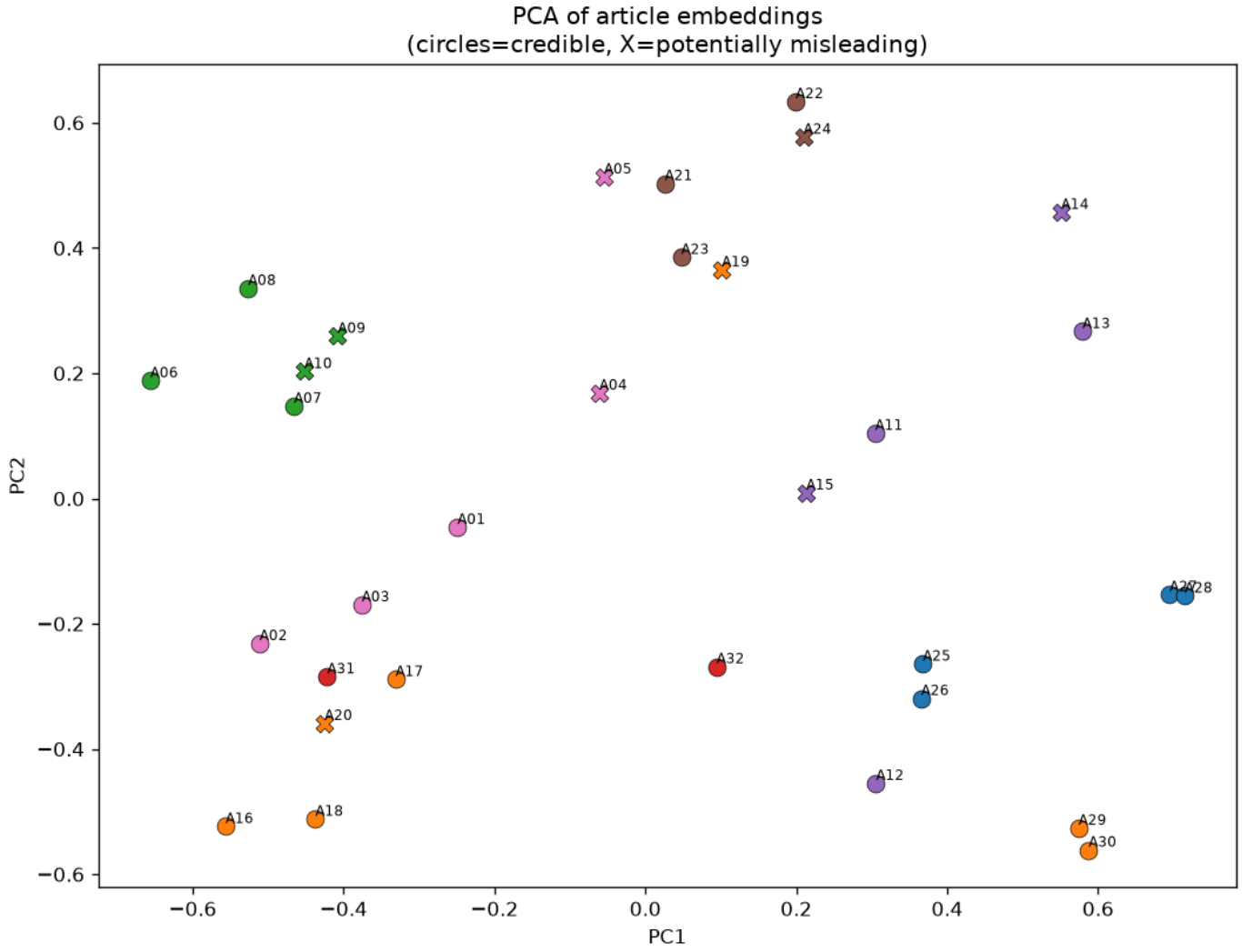
Tutorial – 6

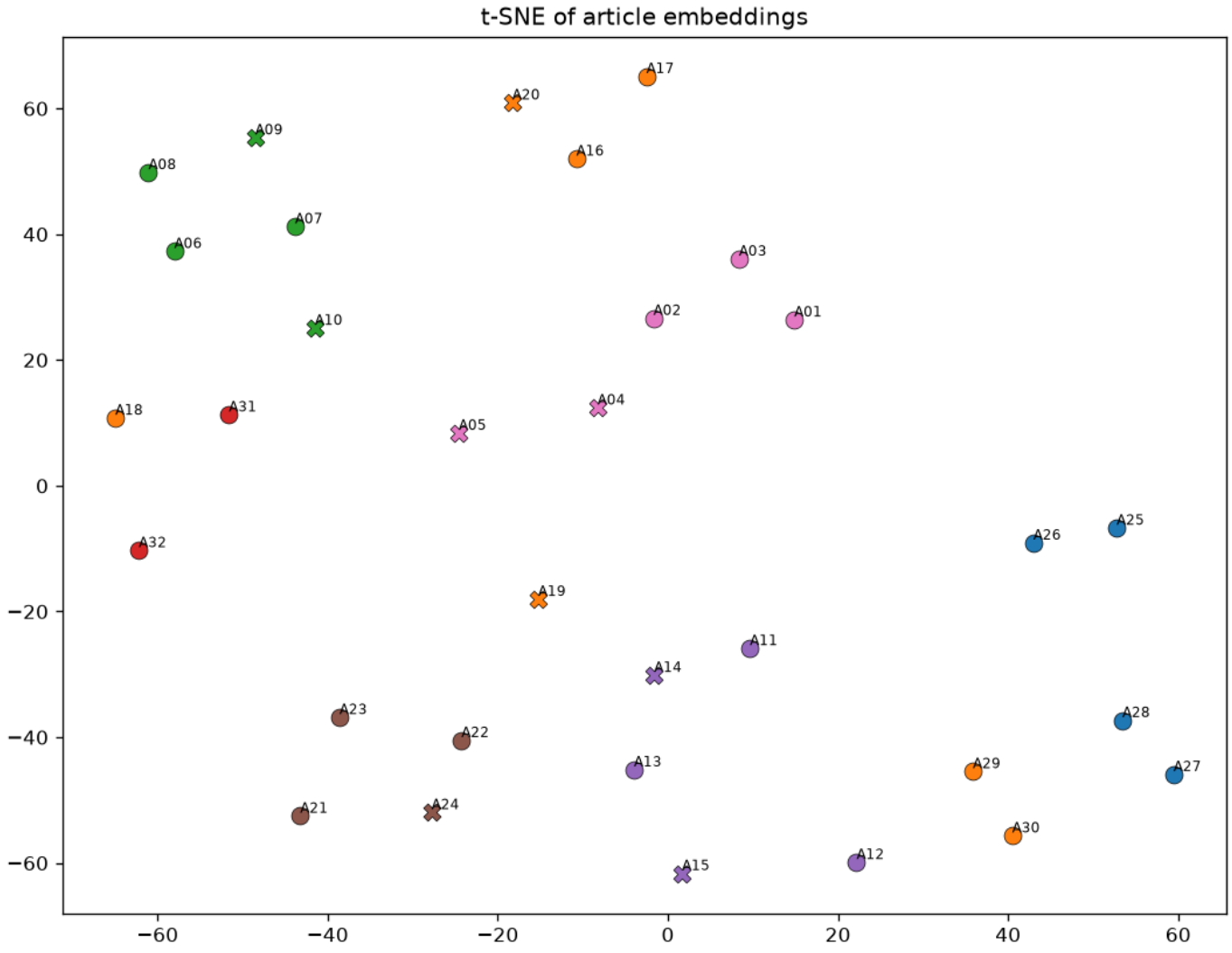

File change.
Rerun
Always rerun
AI-Powered Fake News Detection & Intelligent News Retrieval
Tutorial 6 · U23AI053 · Predictions flag articles as potentially misleading — they do not prove a story is false.
Semantic Search
Analyse Article
Browse Collection
Metrics
Search news by meaning
Query
electric vehicle battery fires and safety
Ranking method
Learning to Rank
Semantic (GloVe)
Both
Search
Learning-to-Rank
A01 · LTR=1.000 · ✓ likely credible (14%)
EV makers publish battery fire safety data after regulator review
reuters.com · Technology · 2024-03-12


File change.
Rerun
Always rerun
A03 · LTR=0.999 · ✓ likely credible (15%)
Peer-reviewed study maps lithium supply chains for electric vehicles
nature.com · Technology · 2024-01-18
Show text
A04 · LTR=0.995 · ⚠ potentially misleading (86%)
Secret study proves every EV battery explodes within two years
viralbuzz.biz · Technology · 2024-04-01
Show text
A02 · LTR=0.989 · ✓ likely credible (12%)
Researchers demonstrate solid-state battery prototype with longer range
bbc.com · Technology · 2024-05-02
Show text
A16 · LTR=0.003 · ✓ likely credible (14%)
Satellite data show continued rise in global surface temperatures
nasa.gov · Climate · 2024-01-10
Show text


Deploy
Verdict: POTENTIALLY MISLEADING
Confidence 84%
0% → likely credible · 100% → potentially misleading
Model votes
MultinomialNB: 56%
LogReg-TFIDF: 68%
RandomForest: 93%
Style-LogReg: 99%
Engineered features
{
"n_tokens" : 40
"n_sentences" : 5
"sensational_ratio" : 0.05
"exclamations" : 0
"questions" : 0
"uppercase_ratio" : 0.012
"avg_word_len" : 5.371
"source_credibility" : 0.12
"tld_suspicious" : 1
"anonymous_cues" : 1
}
Important: Model predictions alone cannot establish whether a news story is true. Treat this as a triage signal for review.


Fake-news classifiers on held-out articles
Accuracy
F1
Score
1.0
0.8
0.6
0.4
0.2
0.0
MultinomialNB
LogReg-TFIDF
RandomForest
Style-LogReg
Ensemble


PCA of article embeddings
(circles=credible, X=potentially misleading)
PC1
PC2
A22
A24
A05
A21
A14
A23
A19
A08
A13
A09
A10
A06
A04
A07
A11
A15
A01
A03
A28
A02
A32
A25
A31
A17
A26
A20
A12
A16
A18
A29
A30
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6


t-SNE of article embeddings
A17
A20
A09
A16
A08
A07
A06
A03
A10
A02
A01
A18
A31
A04
A05
A32
A25
A26
A19
A11
A14
A23
A28
A22
A13
A29
A27
A21
A24
A30
A15
A12
60
40
20
0
−20
−40
−60
−60
−40
−20
0
20
40
60